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Chemistry-related interpretive and reporting errors in geopolymer research: A critical diagnostic review

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ABSTRACT

This purposive critical diagnostic review examines chemistry-related interpretive and reporting risks in geopolymer and alkali-activated materials research. Six domains organize the analysis: terminology and definition errors; precursor chemistry and reactivity ambiguities; activator chemistry reporting and interpretation errors; misuse of composition ratios and chemical variables; mechanism interpretation errors; and errors in linking chemistry to phase assemblage, microstructure, and property. Twelve claim-specific assessments and a comparison with six selected reviews illustrate how mechanism and structure–property interpretations depend on adequate system and variable specification. The proposed framework distinguishes observation strength, causal inference strength, and transferability range as three partly independent dimensions. It pairs diagnostic reading criteria with reporting recommendations and a hypothetical formulation-accounting example. The selected evidence supports bounded interpretation rather than field-wide error frequencies, validated study-quality scores, or independent validation of the framework. Explicit system boundaries, reconstructable chemical information, and evidence-proportional claims are proposed to support reproducibility and chemically defensible comparison.

1. Introduction

Geopolymer research spans diverse precursors, activators, curing routes, and performance targets. Chemically non-equivalent systems can nevertheless be described in apparently uniform language, making informative individual studies difficult to compare mechanistically. Foundational reviews establish the chemical diversity and characterization challenges of alkali-activated binders [1], while methodology-focused reviews address definitions and mix-design methodology [2]. The present review examines how system definition, reaction interpretation, variable reporting, and analytical inference affect comparisons across that diversity.

Terms such as geopolymer, alkali-activated material, and inorganic polymer have overlapping meanings. Thapa and Waldmann discuss their boundaries and the distinction between calcium-rich and low-calcium alkali-activated systems [3]. Wang et al. distinguish alkali-aluminosilicate and silico-aluminophosphate systems by reaction environment and gel chemistry [4]. Grouping distinct systems without chemical qualification can obscure the meaning of variables and the scope of mechanism and structure–property claims.

Comparative review work distinguishes alkali-aluminosilicate and silico-aluminophosphate systems by activation route, pH regime, gel chemistry, and aluminum coordination [4]. Matsimbe et al. [2] highlight variability in precursor properties and note that the reviewed mix-design methods do not include all alkaline activator, precursor, and aggregate parameters. These concerns motivate a chemistry-specific assessment of reconstructability and interpretive support.

The objective is to identify interpretive and reporting risks affecting system definition, reaction interpretation, variable accounting, and chemistry-based comparison, without estimating field-wide prevalence. Relative to the six reviews compared in Section 3, the proposed contribution is a diagnostic reorganization of those risks. The analysis follows six domains: terminology and definition; precursor chemistry and reactivity; activator chemistry; composition ratios and chemical variables; mechanism interpretation; and chemistry–phase–microstructure–property inference. Their dependencies guide the corrective framework proposed in Section 10.

An error is chemistry-related here when it affects (i) the system family assumed, (ii) the reaction pathway inferred from evidence, or (iii) the formulation variables that can be chemically reconstructed. These tests distinguish the diagnostic focus from broader weaknesses such as statistical underpowering, incomplete replication, or general reporting omissions.

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2. Methodology and review scope

This is a purposive critical diagnostic review. Its reported discovery routes and selection principles are described below, but the retained records do not provide a complete database-specific audit trail of exact queries, search dates, retrieved records, duplicate removal and full-text exclusions. The literature search is not presented as independently reproducible. Source identity and claim support were assessed separately during revision; this verification does not establish exhaustive search coverage. The diagnostic focus responds to the need for clearer definitional and methodological standardization across chemically diverse systems [2]. The review does not estimate field-wide error prevalence or claim formal compliance with systematic or scoping review reporting frameworks.

2.1. Search strategy and source logic

The literature base was assembled through structured iterative searches in Web of Science, Scopus, and Google Scholar, supplemented by citation chasing from foundational reviews, mechanistic studies, comparative analyses, and representative experimental articles. Search terms were organized around four conceptual clusters and combined through Boolean operators:

Core terms: *geopolymer*, alkali-activated*, inorganic polymer*, alkali activated material (AAM)*

Chemistry focus: *activator, precursor, mechanism, reaction pathway, gel structure, Si/Al ratio, Na/Al ratio*

Characterization: X-ray diffraction (*XRD), Fourier-transform infrared spectroscopy (FTIR), nuclear magnetic resonance (NMR), scanning electron microscopy (SEM), transmission electron microscopy (TM), synchrotron, molecular dynamics (MD), density functional theory (DFT)*

Reporting and quality focus: *reproducibility, comparability, standardization, reporting, methodology*

Targeted supplementary searches were conducted for solid-state nuclear magnetic resonance (NMR) studies of aluminum coordination, operando synchrotron and pair distribution function (PDF) analyses, computational modelling of geopolymerisation (MD, DFT, ReaxFF, thermodynamic modelling), and The International Union of Laboratories and Experts in Construction Materials, Systems and Structures (RILEM) technical committee (TC) outputs (TC 224-AAM, TC 247-DTA, TC 281-CCC, TC 283-CAM). Reference lists of major comparative reviews were screened for additional candidate sources.

Targeted AI-assisted triage was used as a supplementary discovery tool during the final update phase. Elicit (Elicit Research, accessed August 2026) supplied candidate references for author review and did not make inclusion or exclusion decisions. Bibliographic identity and passage-level claim support were assessed separately during revision, and the audit records the verification basis and remaining limitations for individual sources. Verification provenance is not treated as evidence of the database or tool through which a source was originally discovered. A separate canonical-DOI search-coverage audit was not performed. The reported final literature update covered January 2025 through 16 August 2026 and prioritized studies relevant to chemical reconstruction, mechanism interpretation and evidence-to-claim discipline; subsequent revision-stage verification is distinct from that discovery period. Supplementary Table S3 records the reference-identity, claim-support and publication-status audit, distinguishing explicit author approval of specified uses from historical implementation records and preserving the verification limits for individual sources.

2.2. Inclusion and exclusion principles

Sources were considered for inclusion when they (i) addressed alkali-activated, inorganic polymer, or geopolymer systems with a primary focus on chemistry, mechanism, or characterization; (ii) reported original experimental or computational data, or offered comparative or mechanistic synthesis; (iii) clarified system definition, variable reporting, or evidence-to-claim translation; (iv) appeared in peer-reviewed journals, RILEM technical reports, or recognized scholarly book chapters; and (v) were available in English with accessible full text.

Sources were set aside when they (i) treated geopolymer or alkali-activated chemistry only marginally relative to engineering performance; (ii) did not provide sufficient methodological detail to reconstruct the chemistry under discussion; (iii) duplicated earlier publications without analytical extension; or (iv) lay outside the chemistry-centered focus of this review (for example, structural design or pure life-cycle assessment studies).

2.3. Study selection and analytical set formation

Sources were selected purposively for their relevance to system definition, chemical reconstruction, mechanism interpretation and evidence-to-claim relationships. The resulting set was used to examine bounded claims and informative contrasts; it was not intended to enumerate the field or estimate error prevalence. A complete numerical screening and exclusion ledger is not available in the retained record.

More specifically, studies were included when they (i) explicitly distinguished chemically non-equivalent system families, (ii) reported activator or precursor chemistry with enough detail for reconstruction, (iii) exposed ratio-accounting, water-accounting, or calculation-basis ambiguities, (iv) offered a chemically informative contrast between alkali, phosphate, calcium-bearing, or impurity-rich systems, or (v) illustrated how analytical observation was translated into gel, phase, mechanism, or property claims with a traceable evidence chain. Studies meeting none of these criteria were excluded from the analytical set even when otherwise relevant to geopolymer research.

The analytical core includes foundational chemistry reviews [1], comparative system-family analyses [4], experimental and modelling studies [5], methodology-oriented reviews [2], and studies of alkali-aluminosilicate, calcium-bearing, and silico-aluminophosphate systems. Selected characterization, computational, mechanistic, and reporting studies published through 16 August 2026 inform the evidence discussion. Sections 4–9 examine specific claims from this purposively assembled set; it is not a population sample of the field.

2.4. Transparency statement and scope limits

This review is diagnostic rather than prevalence-focused. It identifies, classifies, and illustrates chemistry-related interpretive and reporting risks but does not estimate their frequency across the geopolymer literature. Selected studies are cited to examine bounded claims, interpretive risks and chemically informative contrasts, not to quantify error occurrence. Where the analysis draws on a single study to illustrate a particular error mode, that example is meant as a clarifying instance, not as a claim about how often the error appears. In this revision, an interpretive error pattern is described as recurrent only when the same specified overextension is documented in work from at least two independent research groups, or explicitly documented in a systematic review or a RILEM interlaboratory study. Otherwise, it is described as an identified interpretive risk or a bounded source-specific limitation. This terminology does not imply a prevalence estimate.

The review also adopts a clear positioning relative to system family. It does not assume that all materials commonly discussed under the geopolymer label are chemically interchangeable. Alkali-aluminosilicate systems, silico-aluminophosphate systems, calcium-bearing alkali-activated binders, impurity-rich industrial by-product systems, and low-impurity model precursor systems count as chemically distinct families that may overlap in practical interest while still differing in reaction environment, product assemblage, and relevant explanatory variables. Cross-system comparisons are made only when the chemistry of each side is sufficiently defined to permit fair contrast, and limits to such comparison are stated where they apply.

Three further boundaries shape the present scope. First, the review focuses on chemistry-related interpretive and reporting errors and does not attempt to re-evaluate all methodological weaknesses of geopolymer studies more broadly; statistical design, replicate structure, and long-term durability testing are addressed only where they intersect with chemistry-specific interpretation. Where a chemistry-to-property causal interpretation is proposed, claim-proportionate reporting of independent replication, variability, quantitative uncertainty and analytical quality control is within scope. This review does not prescribe a universal replicate count, conduct a retrospective statistical reanalysis, or provide a complete general experimental-design standard. Second, the review is constructive in intent, and each diagnosis is paired with a chemistry-focused corrective framework (Section 10).

The six domains are separated here for analytical rather than statistical reasons. Each corresponds to a different epistemic function in the production of chemical interpretation: terminology and definition errors are classification errors; precursor, activator, and ratio-related weaknesses are reconstruction and variable-accounting errors; mechanism interpretation errors are inference errors; and chemistry-structure-property overextensions are evidence-proportionality errors. Precursor-, activator-, and ratio-related errors are retained as separate domains because each concerns a different material or formulation component and needs its own reporting and reconstruction discipline. The domains are then interlinked, but not redundant.

The six domains provide the conceptual organization of this diagnostic review. During revision, their boundaries were clarified against claim-specific source evidence. The eight initially question-specific entries in Table 1 were reassessed and extended to twelve entries using the five diagnostic fields described in Section 9.1. The unit of assessment is the specified chemical question or interpretation within its reported system and evidence base. This revision-stage assessment does not establish that the original search and screening process used a pre-specified claim-coding protocol. Bibliographic identity checking and citation-scope auditing are separate from the domain assessment.

For the revised Table 1 assessment, the primary domain is assigned to the earliest identifiable source of ambiguity in the specified claim: system classification, precursor or activator definition, ratio accounting, mechanism inference, or chemistry-to-property inference. Downstream consequences are recorded as linked secondary domains in the audit and are not counted as separate independent occurrences unless the source makes a distinct claim requiring its own evidence assessment. The rule concerns the dependency within the claim, not the order of the manuscript sections. For positive examples without a diagnosed overextension, the domain identifies the question assessed rather than an error committed by the source. Its application is limited to the Table 1 entries assessed during revision; no retrospective application to the entire original literature set is claimed.

Fig. 1 summarizes the reported purposive selection principles and distinguishes revision-stage verification from original source discovery. It is a conceptual summary, not a numerical screening flow or evidence that the database searches can be repeated from a complete retained search log.

3. Positioning relative to prior reviews

Table 2 compares six selected reviews by their stated chemistry and system focus, methodological or reporting emphasis, and treatment of a chemistry-specific error taxonomy. These are the dimensions assessed in the present comparison, not a reconstruction of a documented original comparator-selection procedure. A complete comparator-selection trail and an exhaustive accounting of related reporting frameworks are not available.

Within these six reviewed comparators, a structured chemistry-specific error taxonomy of the form proposed here was not identified. The present contribution is a diagnostic reorganization that links adequate system and variable specification to mechanism inference and chemistry–structure–property interpretation, distinguishes observation strength, causal inference and transferability, and proposes claim-specific reading criteria and a corrective reporting framework. The comparison supports this contribution relative to the six selected reviews; it does not establish unique novelty across the entire literature or the absence of related frameworks.

The present review does not treat field diversity itself as an error. The error arises when a claim is framed as if it were portable across system families, despite resting on chemistry that remains locally bounded, incompletely defined, or inferentially overextended. Diversity itself is expected; the difficulty arises when findings are transferred across chemically non-equivalent contexts without qualification.

4. Terminology and definition errors

Fig. 2 depicts logical dependencies among specification and interpretation domains. An arrow from X to Y means that interpretation at Y depends on adequate specification at X. Mechanism and chemistry–structure–property explanations therefore depend on upstream system, precursor, activator, and composition definitions through the depicted paths. Omitted direct dependencies do not imply independence; precursor and activator domains are mutually conditioned. The diagram is conceptual, not a temporal sequence, causal error-propagation model, or empirically coded literature network.

Overlapping use of geopolymer, alkali-activated material, inorganic polymer, and related labels can obscure chemical distinctions. A systematic review of geopolymer methodologies links the lack of a common definition and methodology to inconsistency and confusion across disciplines [2]. Ekinci et al. [19] discuss terminology alongside standardization, production and long-term performance priorities. Without explicit boundaries, a shared label can imply comparability of mechanisms, variables, and phase evolution that has not been established.

Thapa and Waldmann [3] report disagreement over whether “geopolymer” can be considered a subset of “alkali activated binders.” Their Fig. 1 nevertheless places geopolymers within the broader alkali-activated materials family, and their conclusions note widespread use of the term for low-calcium or calcium-free systems. They also report Davidovits’ objection to alkaline cement and alkali-activated cement because Portland cement itself hardens under alkaline conditions. The account documents the disagreement while keeping its own classification position.

A second layer of ambiguity appears when the field moves beyond conventional alkali-aluminosilicate systems. Wang et al. [4] show that phosphate-activated aluminosilicate systems have been described through a wide range of partly overlapping labels, including aluminosilicate phosphate cement, phosphoric acid-based geopolymer, phosphate-based geopolymer, acid-based geopolymers, and phosphate-bonded materials. In response, they adopt the broader label “modern geopolymer” and then subdivide this space into alkali-aluminosilicate (AAS) and silico-aluminophosphate (SAP) geopolymers. That move is useful as a comparative device, but it also highlights the extent to which attractive umbrella terminology can conceal substantial differences in activation route, reaction environment, gel chemistry, and aluminum coordination. In their comparative review, Wang et al. [4] associate calcium-free AAS systems with sodium aluminosilicate hydrate (N-A-S-H) gels containing predominantly four-coordinated aluminum and describe SAP systems in terms of silico-aluminophosphate (S-A-P) gels with predominantly six-coordinated aluminum. They also discuss calcium (alumino)silicate hydrate (C-(A)-S-H) formation in calcium-containing AAS systems and its possible coexistence with N-A-S-H in slag–low-calcium-fly-ash blends. These NMR-based coordination assignments remain subject to spectral interpretation and phase attribution and should not be treated as universal fingerprints of the whole binder; a composition-specific NMR example is discussed in Section 6. Such distinctions indicate that nominally related systems may remain chemically non-equivalent even

Table 1
Claim-specific evidence and diagnostic limits in selected geopolymer studies. Five fields distinguish the source's reported claim from the present review's assessment and defensible interpretation. Positive bounded examples are retained without attributing an unestablished error; source passage and review-section locations are identified separately.

Source and passage	Reported claim	Evidence used	Interpretive limitation	Why overextended	Defensible alternative
Fernández-Jiménez & Palomo [6] pp. 2259–2265; discussed in Section 5	Chemical, mineralogical and physical characteristics are relevant to the potential reactivity of type F fly ash.	Complementary precursor analyses and strength testing.	Solution dose is adjusted to normal consistency; the comparison does not isolate ash effects at a constant solution/ash ratio.	No source-specific overextension established for this assessed claim.	Interpret precursor reactivity with the measured characteristics and the reported solution dose.
Weng & Sagoe-Crentsil [5] pp. 2998–3006; discussed in Sections 6 and 8	A dissolution–hydrolysis–condensation interpretation is proposed for low-Si/Al metakaolin activation.	Partial-charge calculations supported by leaching, calorimetric and spectroscopic observations.	The interpretation concerns the tested low-Si/Al metakaolin chemistry; it is not a directly observed universal reaction sequence.	No source-specific overextension established for this assessed claim.	Use the proposed interpretation within its stated chemical scope and distinguish calculations from measured observations.
Wang, Provis & Dai [7] Section 2/Table 1; Sections 3.1–3.2; discussed in Section 6	Soluble aluminum changes sol/gel transition, setting and strength development in metakaolin–phosphate binders.	Solution Al/P, MAP/OPA, pH and water accounting; setting and age-dependent strength.	Solution species and composition co-vary; the early benefit can accompany restricted later strength development.	No source-specific overextension established for this assessed claim.	A bounded composition–response relationship with explicit activator reporting; independent effects of every solution species are not established.
Wang et al. [4] Terminology and Classification; Figs. 3 and 4; discussed in Section 4	SAP and AAS differ in reaction environment, gel chemistry and aluminum coordination.	Secondary comparative synthesis of published experimental and structural evidence.	AAS includes distinct calcium regimes. Coordination assignments remain subject to their spectroscopic basis and uncertainty.	No source-specific overextension established for this assessed claim.	Use the synthesis to bound system comparisons, with calcium scope and coordination uncertainty retained.
Louati et al. [8] Sections 2 and 3; Table 4; discussed in Sections 6 and 7	A Si/P–strength relationship is interpreted through proposed network structure and dissolution/charge-balance explanations.	Strength measurements and XRD, IR and SEM observations.	The measured relationship does not uniquely discriminate the proposed network structure or explanatory mechanism.	The structural and mechanistic specificity exceeds what the reported relationship and observations uniquely establish.	Retain the Si/P–strength observations; state network and dissolution/charge-balance explanations as qualified interpretations.
Koutník et al. [9] Sections 2.3, 3.1–3.3 and 4; discussed in Sections 6 and 7	Total water and Me:Al have response-specific effects in potassium metakaolinite formulations.	Reported composition and tests of rheological, pore and mechanical responses.	The alkali effect on total pore volume is not significant, and the viscosity response depends on water content.	No source-specific overextension established for this assessed claim.	Interpret each formulation–response relationship separately under its reported water conditions.
Lemougna et al. [10] pp. 5280–5286; discussed in Section 5	Iron reactivity depends on the mineral environment, with new ferric environments after activation.	Mössbauer spectroscopy, XRD and SEM/EDS.	The precise structural location of the new ferric environments remains unresolved.	No source-specific overextension established for this assessed claim.	Retain mineral-dependent iron participation; network and amorphous-gel locations remain alternatives.
Burduhos Nergis et al. [11] Section 3.2, Fig. 6; methods, Section 2.2; discussed in Section 9	Specific N-A-S-H, C-S-H and C-A-S-H contributions are assigned to CAMK thermal responses.	TG-DTA response ranges and XRD phase observations.	Thermal ranges overlap; independent confirmation of those specific gel identities is not presented in the audited evidence.	Specific gel identities exceed the discrimination provided by overlapping thermal features. The anhydrite observation is a separate usable result.	Report thermal responses as observations; treat particular gel contributions as interpretations requiring independent confirmation.
Fernández-Jiménez et al. [12] Introduction p. 1204; Fig. 7 p. 1208; discussed in Section 8	The activation model is proposed as independent of fly-ash characteristics and activator type/concentration.	Main series: one class F ash, 8 M NaOH, 85 °C; reaction degree and microscopy. An additional silicate example is shown.	The principal series and extra example do not systematically test independence across precursor and activator conditions.	General independence is broader than the conditions systematically examined; this does not negate the descriptive sequence.	Use the descriptive model under reported conditions; broader independence remains a hypothesis requiring cross-condition tests.
Serag et al. [13] Results and discussions, after Fig. 15; discussed in Section 9	C-A-S-H and N-A-S-H are identified from SEM morphology and EDAX elemental ratios and distributions.	SEM morphology and EDAX ratios; the XRD discussion refers to earlier literature.	Element ratios and morphology do not uniquely identify gel structure; independent gel-specific confirmation is not presented in the audited evidence.	Definite gel-identification language exceeds the structural discrimination supplied by the reported measurements.	The ratios and morphology are consistent with candidate gel environments; specific identities require complementary gel-sensitive evidence.

(*continued on next page*)

Table 1 (*continued*)

Source and passage	Reported claim	Evidence used	Interpretive limitation	Why overextended	Defensible alternative
El Fadili et al. [14] Sections 3.2 and 3.4–3.6, pp. 6–11; Fig. 14; discussed in Section 9	Sugar-lime replacement increases mixed-gel production and shifts hydration from diffusion to interfacial reaction control, explaining strength gain.	Reported blend compositions, strength, XRD, FTIR, SEM/EDS and TGA/DTG.	Ca, Si, Al and P inputs co-vary; gel contributions are not separately quantified, and kinetic control is not experimentally discriminated.	The proposed gel-specific and kinetic explanation is more specific than the evidence resolves. A mechanism schematic does not establish a rate-controlling step.	Retain the tested blend–strength relationship; qualify calcium-mediated gel contributions and kinetic control. No independent sulfate effect is established.
Ben ali et al. [15] Sections 4.2–4.5, pp. 27021–27023; Tables 1, 3 and 5; discussed in Section 9	Hybrid N,K-A-S-H/C,M-(A)-S-H products and a dense GP30 matrix explain improved strength.	Bulk oxide/mix data, strength and porosity; XRD, FTIR and SEM/EDS.	Oxide inputs and ratios co-vary; the proposed gel assemblage and its specific strength contribution are not separately resolved.	Broad diffraction/band features and elemental co-location do not discriminate the proposed hybrid-gel explanation. Porosity decreases beyond the strength optimum.	Retain formulation-specific strength and porosity trends; keep gel/property attribution conditional. Sulfate and impurity effects are not independently isolated.

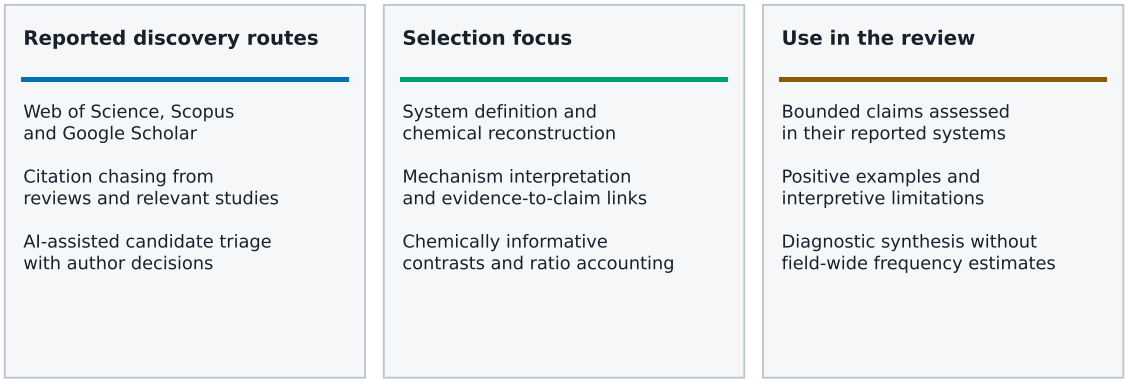


Fig. 1. Purposive selection principles. Conceptual summary of reported discovery routes, selection focus and diagnostic source use. Revision-stage bibliographic and passage-level verification is separate from original discovery and limited to the recorded audit scope. Complete numerical screening and exclusion records are unavailable; no screening counts or search reproducibility are claimed.

when discussed under a common geopolymer banner.

Wang et al. [4] describe the extension of geopolymer terminology from relatively low-impurity alkali-activated aluminosilicates to phosphate-activated systems on the basis of similarities in condensation. Provis and Bernal [1] trace the term to Davidovits' metakaolin-based binders and place geopolymers, commonly associated with low-calcium systems, within alkali-activated materials. The shared name thus encompasses only partly shared chemical identities.

Terminology can obscure whether a binder is a low-calcium alkali-aluminosilicate network, a calcium-bearing mixed-gel system, or an acid-activated silico-aluminophosphate system. It can also make Si/Al, Na/Al, activator concentration, and curing responses appear comparable across unlike environments. Naming a system may implicitly import a reaction narrative or gel expectation from another family, so terminology frames interpretation as well as describing the material.

Umbrella terms should be accompanied by chemically specific system definitions. Alkali-activated material can describe a broad family, provided that low-calcium aluminosilicates, calcium-bearing binders, and phosphate-activated systems remain distinguishable. Geopolymer likewise requires qualification when the formulation departs from the lower-impurity alkali-aluminosilicate model. Classification should not substitute for chemical equivalence.

The potential mismatch between nominal terminology and chemically valid system boundaries is summarized in Table 3.

Table 3 summarizes the terminology distinctions used in this review. Mohan Kumar et al. discuss low- and high-calcium binders and alkali- versus phosphate-activated systems [17], while Fang et al. compare reaction evolution in metakaolin, blast-furnace slag and their blends [18].

5. Precursor chemistry and reactivity ambiguities

Precursor identity should not be conflated with precursor reactivity. Broad labels such as fly ash, slag, metakaolin, halloysite, or clay do not specify a unique reactive input. The type F fly ashes studied by Fernández-Jiménez and Palomo illustrate differences in reactive silica, vitreous content, and particle size [6], while phosphoric-acid activation of illito-kaolinitic clay provides a separate example of mineralogically differentiated precursor chemistry [8]. Serag et al. [13] compared precursor and solid-activator combinations at a fixed total activator dose; activator identity and component proportions varied. Their results concern those combinations rather than an independently varied total-dose effect. C–A–S–H and N–A–S–H assignments in that source rely on SEM morphology and EDAX elemental ratios and distributions; independent gel-specific structural confirmation was not reported.

Potential precursor reactivity and realised reaction extent require different evidence. In Part I, Fernández-Jiménez et al. quantified the vitreous-phase content and composition of two Class F fly ashes using selective dissolution, Rietveld analysis, and solid-state NMR [20]. Part II evaluated reaction degree after alkaline activation using complementary

Table 2
Positioning of the present review relative to six selected reviews in geopolymer and alkali-activated materials research.

Review	Primary focus	Chemistry-specific error taxonomy?	System-boundary discipline?	Main distinction from this review
Provis & Bernal (2014) [1]	Chemistry of alkaline activators, reaction products, setting, and durability in high- and low-calcium alkali-activated binders	No — broad chemistry overview without explicit error classification	Explicit — high- and low-calcium reaction pathways and gel chemistries are distinguished	The present review treats interpretive and reporting errors as the primary analytical object rather than as ancillary observations
Wang et al. (2019) [4]	Comparative chemistry of silico-aluminophosphate (SAP) versus alkali-aluminosilicate (AAS) geopolymers	No — comparative system synthesis without error taxonomy	Yes — explicit SAP/AAS distinction with gel-chemistry and Al-coordination contrasts	The present review extends system-boundary discipline beyond SAP/AAS to terminology, activator, precursor, ratio, and evidence-hierarchy errors across multiple system families
Matsimbe et al. (2022) [2]	Systematic review of geopolymer definitions and mix-design methodologies	Partial — gaps in mix-design variables identified, not organized as a chemistry-centered taxonomy	Limited — formulation and design emphasis dominates over chemistry-class boundaries	The present review reframes reporting weaknesses as chemistry-centered interpretive errors and links them to mechanism, comparison, and structure–property claims
Siyal et al. (2024) [16]	Synthesis kinetics and formation mechanism of geopolymers	No — mechanism-stage discussion without an explicit error classification	Limited — kinetic and mechanistic emphasis, with explicit contrasts between some precursor–activator systems	The present review treats mechanism interpretation as one of six interlinked chemistry-related error domains rather than as an isolated synthesis question
Mohan Kumar et al. (2025) [17]	Comprehensive review of geopolymer chemistry and composition	No — broad chemistry overview without explicit error classification	Explicit — low/high-calcium and alkali/phosphate system distinctions	The present review focuses specifically on chemistry-related interpretive errors and their dependencies on system and variable specification rather than providing a compositional overview
Fang et al. (2025) [18]	Critical review of reaction evolution and mechanism in alkali-activated metakaolin, slag and metakaolin/slag systems	Partial — relates mechanistic uncertainty to multicomponent systems, concurrent reactions and characterization limits; no structured error taxonomy	Yes — explicit focus on metakaolin/BFS system boundaries	The present review extends the diagnostic approach to the full spectrum of chemistry-related error types across multiple system families, including mechanism interpretation and evidence hierarchy

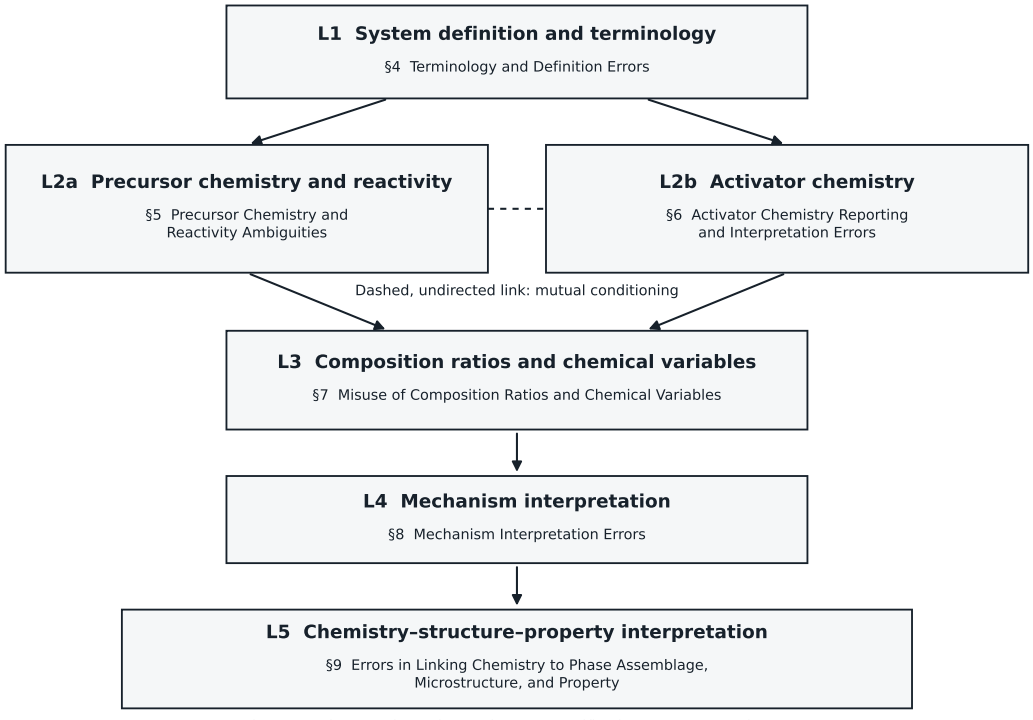


Fig. 2. Logical dependencies among domains that require specification or interpretation. L1, system definition and terminology; L2a, precursor chemistry and reactivity; L2b, activator chemistry; L3, composition ratios and chemical variables; L4, mechanism interpretation; L5, chemistry–structure–property interpretation. Each node includes its section cross-reference in the reorganized manuscript. An arrow from X to Y indicates that interpretation at Y depends on adequate specification at X, not that errors propagate causally. The dashed, undirected precursor–activator link denotes mutual conditioning. Dependence along the depicted paths is transitive: later interpretations also depend on the adequacy of upstream specifications. Direct dependencies may bypass intermediate layers but are not individually drawn; their omission does not imply independence. No specific skip-layer route is asserted because the twelve-record Table 1 coding does not independently verify such routes.

methods [21]. The framework therefore distinguishes potential reactivity from reaction extent under stated activation conditions.

Table 3
System-boundary and terminology matrix for commonly used terms in geopolymer research.

Term used	Typical intended meaning	Chemically valid scope	Common misuse	Consequence of misuse
Geopolymer	General label for non-OPC inorganic binders formed from aluminosilicate precursors	Should be used only with explicit chemical qualification when system family is clear	Used as a catch-all term for chemically distinct cement-free binders	Makes unlike systems appear directly comparable
Alkali-activated material (AAM)	Broad family of binders produced through alkali activation	Valid umbrella term for chemically diverse alkali-activated subclasses	Used interchangeably with geopolymer as though both denote the same chemistry	Masks diversity in gel chemistry and reaction pathways
Inorganic polymer	Structure-focused description of a network-forming mineral binder	Useful only at a broad conceptual level	Used as a prestige synonym for geopolymer without chemical clarification	Replaces system definition with vague structural language
Alkali-aluminosilicate geopolymer	Low-calcium alkali-activated aluminosilicate system	Appropriate where alkali aluminosilicate gel chemistry is dominant	Extended to calcium-bearing or impurity-rich systems without qualification	Imposes low-calcium logic on mixed-chemistry systems
Silico-aluminophosphate geopolymer	Phosphate-activated aluminosilicate system	Appropriate where phosphate chemistry governs dissolution and network formation	Treated as directly equivalent to alkali geopolymers because both are hardened amorphous binders	Ignores differences in pH regime, gel family, and coordination chemistry
Phosphate-based geopolymer	General label for phosphate-activated geopolymer-like binders	Acceptable only when acid route and chemistry are explicitly defined	Used without clarifying whether the system is SAP-type, aluminum-phosphate-rich, or mixed	Encourages transfer of alkali-derived interpretation to acid-activated systems
Calcium-bearing alkali-activated binder	Alkali-activated binder where calcium materially influences gel assemblage	Appropriate for slag-rich and related systems	Described simply as geopolymer without noting calcium-controlled chemistry	Understates coexistence of different reaction products
One-part geopolymer / one-part AAM	Dry-activator route; “just add water” processing format	Valid as a processing descriptor	Treated as though it defines a unique chemistry	Confuses delivery format with reaction chemistry
Binder	Neutral engineering term for the hardened matrix-forming phase	Broadly valid in engineering description	Used where chemistry-specific identification is needed	Removes distinctions that matter for mechanism and comparison
Hardener	Practical formulation term for the liquid or additive promoting setting	Acceptable only in non-mechanistic formulation language	Used instead of activator in chemistry discussion	Minimizes the role of the liquid phase in dissolution and speciation
Chemically activated material	Very broad label for non-traditional binder formation	Useful only as a high-level umbrella term	Used as though it communicates actual system identity	Provides almost no mechanistic or comparative clarity

Related work on fly ash–dehydroxylated clay blends found that increasing the alkaline solution-to-binder ratio generally increased the measured degree of reaction; because the activator volume changed, NaOH, OH^- and water doses varied together [22].

Material labels can encompass different amorphous fractions, silica and alumina availability, mineral assemblages, impurities, particle sizes, and dissolution behaviour. Naming the precursor alone leaves reactive and persistent fractions unresolved. Type F fly ash comparisons relate alkaline reactivity to reactive silica, vitreous content, and particle size within the tested series [6]. Calorimetry of slag, metakaolin, and two ceramic powders activated with a defined potassium silicate solution provides a formulation-specific reactivity indicator [23], not a universal precursor-suitability criterion.

Phase abundance, condition-specific solubility, local coordination and reaction heat answer different questions. Quantitative XRD, selective dissolution, solution analysis, calorimetry, solid-state NMR and direct dissolution tests should therefore be interpreted by their observable and test conditions, rather than treated as interchangeable measures of reactive content (Supplementary Table S1). Solution analysis may be the readout of a dissolution test, so the two are not automatically independent evidence.

Fernández-Jiménez and Palomo [6] associated the potential alkaline reactivity of type F fly ashes with reactive silica, vitreous content, and particle size. Ashes in the same category differed in those features and in crystalline constituents. Their mortar comparisons used 8 M NaOH, but solution-to-ash ratios were adjusted to normal consistency rather than held constant. The results support precursor-sensitive interpretation under those preparation conditions without isolating precursor effects at an identical liquid dose.

XRD and ^{29}Si MAS NMR in that fly ash study distinguished vitreous and crystalline components, including mullite, quartz, hematite, and minor phases. Similar bulk oxide families need not provide the same reactive input. Bulk Si and Al contents alone do not establish the fractions available under the activation conditions.

Louati et al. [8] studied phosphoric acid-activated illito-kaolinitic clay containing kaolinite and illite with quartz, muscovite, gypsum, hematite, and dolomite in the raw material; some associated phases persisted after calcination. Illite was interpreted as contributing only modestly to geopolymerization, on the basis of semi-quantitative XRD phase estimates, and quartz and hematite persisted after activation; the authors interpreted hematite as not contributing, while the kaolinite-derived fraction showed evidence of reaction. Interpretation of “clay-based geopolymer” must therefore distinguish the precursor’s mineral fractions.

Kaze et al. [24] found that Cameroonian halloysite required calcination at 600 °C to produce reactive meta-halloysite under their synthesis conditions; lower temperatures gave non-reactive behaviour or poor matrix stability. Calcination history and dehydroxylation state belong to the precursor definition, even for a mineral family structurally analogous to kaolinite.

Lemougna et al. [10] used ^{57}Fe Mössbauer spectroscopy, XRD, and SEM to examine volcanic ash activation. Part of the iron-bearing augite reacted, producing ferric environments compatible with distorted tetrahedral or five-coordinated sites. These suggested network incorporation or an amorphous gel-like phase, but the spectra did not uniquely distinguish the alternatives. Impurity participation must consequently be assessed rather than inferred from an oxide list or assumed inertness.

Matsimbe et al. [2] identify precursor variability as a barrier to adoption and standardization. It also limits interpretation: comparing two ashes or clays by material label alone leaves their reactivity-relevant differences unresolved. Characterization must address the selected chemical question.

Wang et al. [4] discuss impurity-related side chemistries in industrial by-products and natural aluminosilicates, including non-geopolymeric binders or mixed-gel environments under particular activation conditions. Shared aluminosilicate components do not establish a common reactive input across slag, fly ash, thermally activated halloysite, and phosphate-treated clay.

For chloride-binding comparisons, Li et al. [25] calculated

vitreous-fraction oxide compositions from XRD and XRF data. Their calculation assumed that only the vitreous fraction was reactive and treated slag as fully vitreous, excluding loss on ignition. These estimates do not establish how much of the vitreous fraction reacts under specified activation conditions. More generally, precursor descriptions should state silica and alumina availability, amorphous or vitreous fraction, persistent crystalline phases, relevant impurities, and thermal activation history where applicable. Missing precursor information can confound attribution of formulation differences to activator choice, ratios, or curing.

El Fadili et al. [26] review alkaline aluminosilicate dissolution as a function of alkalinity, activator identity and concentration, fineness, impurities, leaching temperature, and contact time. The synthesis supports treating dissolution reactivity as condition-dependent, within the systems examined; it is not a universal quantitative dissolution model.

A shared precursor class says little about the reactive input on its own. Because dissolution and transformation also depend on the surrounding solution, the activator has to be defined with the same care.

6. Activator chemistry reporting and interpretation errors

Activator labels do not define a chemically stable reactive environment on their own. In the reporting framework developed here, activator definition therefore includes hydroxide concentration, soluble silicate contribution, cation identity, liquid composition, and water supplied through the activator. These descriptors must be interpreted within the relevant system family: alkali-aluminosilicate and silico-aluminophosphate systems differ in reaction environment and gel chemistry [4].

Two studies using nominally identical activators may differ in silicate modulus, hydroxide concentration, cation, water content, soluble aluminum, or acid loading. These differences affect reconstruction, system classification, and comparison. Matsimbe et al. [2] identify activator type and concentration as important variables while noting the lack of common definition and methodology.

A label such as “NaOH-activated” identifies the alkali cation but does not specify silicate supply, modulus, or mixture water distribution; a generic “alkali-activated” label does not specify even the cation. Koutník et al. [9] held Si:Al constant while varying Me:Al and total water in potassium-activated metakaolinite binders. Effects depended on the response and water level: Me:Al had no significant viscosity effect at total water contents of 40 wt% or higher. Interpreting workability, strength, and pore development requires that formulation detail.

For a single fly ash, Rattanasak and Chindaprasirt found that Si and Al leaching depended on NaOH concentration and solution contact time [27]. Alrefaei et al. found water-to-precursor-dependent superplasticizer effectiveness in one-part fly ash–slag pastes activated with anhydrous sodium metasilicate [28]. Palacios and Puertas reported admixture- and activator-dependent responses in slag: a naphthalene-based admixture in NaOH-activated slag improved paste flowability and reduced the mortar liquid-to-solid ratio [29]. These bounded results require formulation-specific comparison.

Mixed activator systems present a second problem when they are discussed as though they were chemically equivalent to single-component systems. Sodium hydroxide and sodium silicate are often grouped together under general alkaline activation language, yet the literature clearly shows that the presence of soluble silicate changes reaction pathways, gel continuity, and matrix development. Wang et al. [4] note that a reasonable content of silicate species in the activating solution can promote a more continuous alkali-aluminosilicate gel matrix, particularly when the precursor has a relatively low Si/Al ratio. Weng and Sagoe-Crentsil [5] provide a complementary analysis of low-Si/Al, NaOH-activated metakaolin, using partial-charge calculations and experimental observations to relate alkalinity, silicate speciation and possible condensation pathways. Their study did not directly compare hydroxide-only and hydroxide-plus-silicate formulations. The distinction between hydroxide-only and hydroxide-plus-silicate activation should therefore be retained when defining the chemical system. It is a mechanistically relevant system boundary. When that distinction is blurred, the literature risks comparing structurally different activation routes under a single descriptive label. The same reporting problem extends to one-part systems, where the chemical meaning of sodium silicate depends on nominal activator identity and on the source chemistry and solid-form delivery route used to generate the reactive environment [30].

Matsimbe et al. [2] summarize reported ranges of NaOH molarity, Na_2SiO_3/NaOH, SiO_2/Na_2O, Si/Al, Ca/Si, and liquid-to-binder ratio while noting the absence of a harmonized mix-design standard. “Optimal molarity” or “best activator ratio” is interpretable only with the cation, silicate contribution, and calculation basis specified.

Koutník et al. [9] discuss a potassium–sodium viscosity comparison from earlier literature, but their own potassium-based series does not provide a matched cation comparison. Reports should state the cation explicitly; general terms such as “alkaline activator” leave a potentially relevant variable unresolved.

The reporting problem becomes more severe when phosphate-activated systems are drawn into the same discursive space. Comparative review work makes clear that alkali-aluminosilicate and silico-aluminophosphate systems differ not only in activator composition, but in pH environment, dominant gel type, and the chemical environment of aluminum. The NMR-based aluminum-coordination contrast summarized in Section 4 is another system-specific distinction [4]. Goryunova et al. [31] equally distinguish the structural units of alkali-activated geopolymers from those of phosphoric acid-activated systems, listing aluminosilicate building units for the former and phosphate-bearing units such as –Al–O–P–O– and –P–O–Si–O–P–O– for the latter. Critical review work on phosphate-based geopolymers identifies competing structural and formation-mechanism proposals and limitations in the analytical evidence used to distinguish them [32]. These uncertainties reinforce the need to evaluate phosphate-system interpretations against their own chemical and analytical basis. Solid-state ^{27}Al MAS NMR of metakaolin–phosphoric acid geopolymers shows that the coordination proportions are composition-dependent rather than fixed. In this series, the Al/P formulation, added water and reaction extent changed together: as the Al/P molar ratio decreased from 4.0 to 0.8, Al^{IV}–O–P species fell from about 20% of aluminum units to nearly zero, Al^{VI}–O–P rose from about 40% to about 80%, and residual metakaolin aluminum environments persisted where phosphorus was insufficient [33]. In the most acid-rich composition, berlinite was quantified at only about 2%, alongside orthorhombic $AlPO_4$ and Si–P crystalline phases [33]. Coordination statements for silico-aluminophosphate systems should therefore be tied to the composition, reaction extent and NMR evidence of the system examined. Added water ranged from 50 to 140 g per 300 g of metakaolin across the Al/P formulations, while curing was held constant [33]. Water was not varied independently of composition, so the series does not isolate a water-content effect; the fixed curing regime likewise provides no independent test of curing effects. Such differences mean that “activator” is not simply a reagent choice layered onto an otherwise unchanged chemistry. It is part of the system definition itself. Any comparison between alkali and phosphate activation that does not acknowledge this distinction begins from a chemically unstable premise.

Phosphate activation is also underdefined when acid concentration alone is reported as though phosphoric acid loading were sufficient to define the system. Louati et al. [8] report a Si/P-dependent strength trend in phosphoric acid-activated illito-kaolinitic clay and discuss proposed dissolution, bonding and charge-balance explanations. The proposed pathway and its causal contribution to strength are not independently established by the reported XRD, IR and strength observations. Wang, Provis, and Dai [7] prepared metakaolin-based phosphate binders using monoaluminum phosphate and orthophosphoric acid, with activating-solution Al/P ratios of 0, 0.1, and 0.3 at fixed Si/P and

total H_2O/MK ratios. Soluble aluminum accelerated the sol/gel transition and shortened setting time. The Al/P = 0.3 mixture gained early strength but showed little further development between 7 and 28 days, whereas the aluminum-free activator mixture continued to gain strength. The authors attributed the limited later development to restricted subsequent reaction of metakaolin. The result is an early-strength/later-development tradeoff within the tested phosphate system. These examples show that phosphate activation cannot be fully described by naming H_3PO_4 and giving one concentration value. The chemically active species in the solution, and the ratio logic by which acid and precursor are matched, must be considered part of the activator definition.

The term hardener can obscure the liquid phase's roles in dissolution, speciation, oligomer formation, gel chemistry, and charge balance. When mechanism is discussed, practical formulation language should be accompanied by a chemically explicit activator definition.

Apparent agreement or disagreement in property trends can reflect inconsistent activator accounting as well as real chemical differences. Before broad labels such as concentration or ratio are used to compare formulations, the activator itself needs to be reconstructed.

Two activators reported under the same label can differ chemically, and so can their preparation histories. For the sodium hydroxide–silicate solutions examined, Skane et al. identified water → alkali hydroxide → soluble silicate as the more reliable preparation sequence [34]. Their related experimentally validated energy-balance model incorporated thermodynamic modelling to predict temperature profiles, cooling times, and solution stability [35]. The present framework recommends reporting addition sequence, solution temperature, time between preparation and use, and observed precipitation or viscosity changes alongside composition.

Section 10 consolidates the reporting recommendations. Composition ratios, discussed below, are only as reliable as these precursor and activator inputs.

7. Misuse of composition ratios and chemical variables

Nominal composition ratios are not chemically self-sufficient descriptors unless their accounting basis is made explicit. Ratios such as Si/Al, Na/Al, or water/solid condense complex formulation information into compact symbols, but their meaning depends on what is counted and on which chemistry is effectively available for reaction. Within the present framework, numerical similarity in reported ratios is not sufficient to establish chemical equivalence. A related distinction concerns batched composition versus effective chemical environment: geopolymerisation proceeds through dissolution, hydrolysis, condensation, and gel evolution in a medium whose chemistry changes during processing, so batch composition is not automatically identical to the environment actually experienced by the dissolving precursor or the growing gel. Hu et al. [36] adjusted the reported Si/Al ratio by changing the proportions of fly ash, glass powder, and sodium silicate solution, so the ratio series also changed the silica-source combination. Yi et al. [37] explicitly expressed composition through SiO_2/Al_2O_3, Na_2O/Al_2O_3, and H_2O/Na_2O molar ratios, illustrating a case in which the ratio basis must be stated. Conceptual stability as a result requires that authors distinguish what is dosed, what is dissolved, what is structurally incorporated, and what is only inferentially assumed.

Si/Al, Na/Al, H_2O/M_2O, water/solid, and activator/binder describe formulation space, but require explicit calculation bases and system context. Matsimbe et al. [2], citing Nodehi and Taghvaee, report total molar ratio, alkali concentration, and activator ratio among variables linked to performance while noting the lack of a harmonized mix-design standard. Numerical agreement alone cannot establish equivalence.

In alkali-aluminosilicate systems, Si/Al interpretation depends on precursor composition, activator-supplied silicate, precursor reactivity, calcium-bearing phases, and relevant impurities. The AAS/SAP comparison by Wang et al. [4] illustrates distinct reaction environments and network chemistries; the present framework accordingly treats ratio meaning as system-dependent. Identical numbers need not denote equivalent chemical balances.

Ratios are also sometimes treated as independent explanations, although they are derived descriptors. Koutník et al. [9], in optimizing low-viscosity metakaolinite-based geopolymer binders, maintain a constant Si:Al ratio while varying Me:Al ratio and total water content. Increasing total water content reduced viscosity and mechanical performance and increased pore volume and mean pore size. Me:Al effects were response-specific: alkali content did not significantly affect total pore volume, and its viscosity effect was not significant at water contents of 40 wt% or higher. Yet the importance of that study lies not only in identifying sensitive ratios; it lies in demonstrating that the effect of each variable is interpretable only because the surrounding chemistry is tightly controlled and clearly reported. Wang et al. [38] compared strength and SEM observations across Si/Al and Na/Al series and examined mass loss under early sealed and open curing in a separate water series. The latter does not independently establish a free-water effect on both strength and micromorphology. More broadly, ratios do not explain geopolymer systems in isolation; they explain a system within a defined precursor, activator family, cation type, and curing pathway. Where these controls are not reported, numerical ratio labels alone do not establish chemical comparability.

Ratio bases may include bulk precursor, total batch, an estimated reactive fraction, activator solution, or nominal oxide equivalents. Matsimbe et al. [2] discuss the prominence of ratio variables and the absence of a standardized mix-design method. The framework therefore requires each denominator and included component to be stated: "Si/Al" cannot distinguish those accounting choices on its own.

Water-related variables raise the same comparability problem. Weng and Sagoe-Crentsil [5] treated water as a participant in dissolution, hydrolysis and condensation, and monitored changes in free-water-related NMR relaxation during reaction. Their experiments varied NaOH concentration rather than independently testing water dosage, so they do not establish an isolated water-content effect on condensation pathways. Separately, Koutník et al. [9] demonstrate that increasing total mixture water from 30 to 50 wt% reduces viscosity and mechanical performance while increasing pore volume and mean pore size in potassium-activated metakaolinite binders. Their water accounting includes water in the activating components and added water; the specimens received a sealed 60 °C, 4 h initial cure followed by 7 days at 20 °C. These observations indicate that water-related variables cannot be interpreted solely through workability language. Yet in many published discussions, water/solid or related ratios are compared numerically across studies without clarifying whether water is counted as total batch water, free added water, water contained in commercial silicate solutions, or water associated with acid dilution. When the chemical role of water is reduced to a single operational ratio, part of its mechanistic significance disappears from the analysis.

Louati et al. [8] report a compressive-strength optimum with Si/P in phosphoric acid-activated illito-kaolinitic clay. The trend is bounded by that formulation and activation route. It cannot be transferred to alkali-activated metakaolin or fly ash as though Si/P, Si/Al, and Na/Al described the same chemical balance.

An "optimal" ratio is conditional on precursor chemistry, activator composition, curing, and the performance criterion. Matsimbe et al. [2] discuss trial-and-error design and the lack of unified standards. Optimization within a tested formulation does not establish a transferable ratio target.

These problems do not imply that ratio language should be abandoned. On the contrary, ratio-based description remains indispensable in geopolymer chemistry. Ratios work best as context-bound descriptors whose interpretive value depends on explicit chemical definition. At minimum, studies should clarify how each ratio is calculated, which components are included, whether the value refers to total or effectively reactive chemistry [39], and whether cross-study comparison is

restricted to chemically comparable system families. Without such discipline, ratio-based discussion can create a false impression of precision while quietly eroding mechanistic and comparative clarity. The present framework further recommends that, in silico-aluminophosphate systems, studies state whether Al/P is based on total precursor composition, an independently characterized amorphous fraction, or a condition-specific reactive fraction; in this framework, amorphous and reactive bases are not treated as interchangeable. Reactive-basis estimates require the method, activation conditions and uncertainty to be reported. Keppert et al. calculated Al/P using an assumed amorphous Al_2O_3 content of 40.8 wt% of the metakaolin [33]; this identifies their calculation basis without establishing the amount of Al that reacts under every activation condition. The present framework additionally recommends reporting water content and curing history for comparison.

For the hypothetical formulation in Supplementary Table S2, precursor-only Si/Al is 1.0413, whereas total-batch Si/Al is 1.4733; the corresponding total SiO_2/Al_2O_3 is 2.9466. Atomic ratios such as Si/Al and Na/Al differ from oxide molar ratios by stoichiometric factors ($SiO_2/Al_2O_3 = 2 \times Si/Al$; $Na_2O/Al_2O_3 = Na/Al$), so the notation used should always be stated. The conditional reactive-basis value of 1.3477 depends on explicitly assumed, unequal Si and Al accessible fractions. Initial NaOH molarity and batch-normalized NaOH inventory are reported separately and do not determine measured free OH^- or pH. The example also states the water denominator and distinguishes physical solution water from formal oxide-accounting water.

Ratios remain interpretable only together with system identity and calculation logic. Numerical similarity is not chemical equivalence, and the same caution applies to the mechanism claims built on these definitions.

8. Mechanism interpretation errors

A dissolution–condensation–hardening sequence can introduce geopolymerisation, but it is not a complete mechanism. Precursor dissolution, hydrolysis, species evolution, condensation, gel formation, and structural reorganization can overlap and depend on the system considered [5,16,40]. De Silva et al. [41] related aluminium supply to setting and silicon supply to later strength in a metakaolin system; that study does not test a universal sequence across binder families. Mechanism interpretation must address those dependencies after formulation specification.

A related source of interpretive fragility is the tendency to treat indirect diffraction or spectral evidence as though it resolved mechanism on its own. Provis et al. [42] reinterpreted published diffraction and microscopy data and proposed that geopolymeric binders may contain zeolite-like nanocrystals within an amorphous aluminosilicate gel. This structural proposal should be distinguished from direct evidence establishing a unique formation mechanism. Fernández-Jiménez and Palomo [43] compared sodium-based activator formulations and reported associated changes in gel composition and microstructure. Cation-identity comparisons discussed in their introduction derive from earlier literature; their own experimental series does not compare Na and K activators. Mechanism should therefore not be inferred from broad structural resemblance alone, nor transferred across chemically distinct systems without explicit qualification; a defensible mechanism claim must remain bounded by precursor state, activator chemistry, silicate speciation, and the analytical strength actually supporting the proposed pathway [5,40,44]. The remainder of this section examines specific interpretive patterns where this discipline most often breaks down.

Xu and van Deventer [45] describe dissolution, orientation or transportation, and reprecipitation or polycondensation while acknowledging an incompletely resolved hardening mechanism. White et al. [40], in a multiscale simulation study, discuss concurrent dissolution and repolymerization and the difficulty of characterizing each process experimentally. A rigid sequential narrative would obscure that concurrency and uncertainty.

Weng and Sagoe-Crentsil [5] distinguish low-Si/Al metakaolin systems from higher-silicate derivatives. Their partial-charge calculations, interpreted with experimental observations, predict how alkaline silicate speciation may favour smaller oligomers or larger aluminosilicate networks. The interpretation remains bounded by the silicate supply, alkalinity, precursor, and reaction conditions examined.

The AAS/SAP boundaries discussed in Section 4 constrain transfer of mechanisms across activation routes [4]. For phosphoric acid-activated illito-kaolinitic clay, Louati et al. [8] discuss acid dissociation and Al–O–P bonding as a proposed reaction pathway, drawing on earlier literature, rather than as a directly resolved sequence. Applying an alkali-aluminosilicate narrative without accounting for that reaction environment would overextend the analogy.

Kinetics describes how quickly a system evolves, not necessarily the species or pathway involved. Siyal et al. [16] review calorimetry-based and Avrami-type approaches while identifying technique and model limitations. White et al. [40] discuss concurrent-process characterization and the computational demands of a heterogeneous material across scales. A calorimetric peak, setting response, or fitted exponent can inform a mechanism proposal but cannot establish it alone.

Egnaczyk, Quinn, and Wagner [46] studied a defined metakaolin geopolymer formulation using isothermal calorimetry, rheology, and $^{29}Si/^{27}Al$ solid-state NMR, acquiring the NMR spectra at −17 °C to quench the reaction at defined extents of reaction. By coupling kinetic rate data with direct structural characterization at defined degrees of reaction, the study avoids the inferential shortcut of reading mechanism from rate alone. The authors explicitly state that other elements, including Ca, K, Fe, and Mg, must be incorporated to confirm the generalizability of the proposed relationship beyond their sodium–metakaolin model. Aligning kinetic and mechanistic evidence within defined system boundaries in this way gives a useful model for reporting in geopolymerisation mechanism studies.

Descriptive microstructural sequences create a further risk when they are read as universally valid mechanistic templates. Fernández-Jiménez et al. [12] proposed a descriptive model for the microstructural development of alkali-activated fly ash cement, distinguishing dissolution and polymerization stages and showing how reaction products progressively compact around partially reacted spheres. The authors explicitly proposed the model as generally applicable to fly-ash alkali activation, independently of fly ash characteristics and activator type or concentration. Their principal experimental series used one class F fly ash, 8 M NaOH and curing at 85 °C; an additional sodium-silicate-activated example was presented with reference to earlier work. These observations support the descriptive sequence but do not systematically establish its independence from precursor and activator conditions. Its proposed generality within fly-ash systems should also be distinguished from applicability to all geopolymer chemistries. Even within the same broad precursor family, differences in vitreous content, reactive silica availability, activator chemistry, and curing regime can alter which processes dominate and how strongly they affect final structure. Descriptive models are most useful when kept within their evidential domain. They become problematic when converted into generalized reaction doctrine that silently erases chemical diversity across systems.

Dissolution, hydrolysis, condensation, gelation, oligomerization, nucleation, and hardening denote different processes whose roles depend on the system. Source-specific examples include thermal transformation of corundum-containing, potassium-activated metakaolin [47] and multiscale characterization of slag–coal gangue composites by XRD, FTIR, solid-state NMR, electron microscopy, and nanoindentation [48]. These provide local formulation–structure–property evidence; keeping mechanism language within its evidential scope is the present review's synthesis.

Recent computational work has begun to address some of these

interpretive fragilities at a finer level. Yang et al. [49] used ab initio molecular dynamics at 350 K to examine selected aqueous silicate and aluminate reactions. The modeled pathways included metastable pentacoordinate intermediates in selected reactions and lower calculated barriers when an Al–O tetrahedron acted as the attacked tetrahedron. Complementary DFT and ReaxFF molecular dynamics calculations examine how cation identity, selected monomeric and dimeric aluminosilicate species, and graphene surface functionalization affect interfacial adsorption energies [50]. These calculations concern adsorption at the modeled interfaces and do not directly establish a bulk-binder dissolution or network-formation pathway. Such atomistic evidence does not replace experimental mechanistic inference, but it strengthens the case that mechanism claims must remain bounded by system-specific chemistry rather than transferred across unlike system families by analogy.

9. Errors in linking chemistry to phase assemblage, microstructure, and property

XRD amorphous humps, FTIR band shifts, TG-DTA events, and SEM morphology inform material development. They do not independently establish precise bond-formation routes, unique gel identities, or a complete mechanism. Interpretation becomes overextended when a conclusion is more specific than the signal can discriminate.

One interpretive risk is the conflation of gel type with final phase assemblage. Comparative review work has made clear that geopolymer systems may contain different primary gels, secondary reaction products, persistent precursor phases, and impurity-driven side phases depending on precursor chemistry and activator route. Wang et al. [4] also review impurity-related side chemistries that can contribute additional phases to the binder assemblage. Yet many individual studies move from one or two analytical signatures directly to statements such as “the geopolymer gel formed was X” or “the structure transformed into Y,” without sufficiently separating broad network indication from full phase resolution. Once gel identity is read directly from limited data, phase assemblage and reaction mechanism are easily merged into a single unsupported conclusion. Jin et al. [51] reported precursor-dependent crystalline phase assemblages and activator-dependent changes in residual and reaction-product signals in separately studied slag and fly ash systems. This provides a system-specific example of why a primary-gel label does not describe the full phase assemblage. This caution is especially relevant in blended slag–metakaolin systems, where binder evolution is not adequately captured by singular gel language and where structural development remains sensitive to both precursor combination and activation route [52].

Overinterpretation of XRD and FTIR patterns is another identified interpretive risk. Burduhos Nergis et al. [11] used XRD and TG-DTA to examine phase evolution and thermal behaviour in phosphate-activated coal ash- and metakaolin-based formulations, including mixtures with mine tailings. In the mine-tailings-containing samples, they reported the disappearance of the characteristic anhydrite diffraction signal and interpreted it as dissolution of $CaSO_4$ in the presence of phosphoric acid. That approach is reasonable when kept at the level of phase indication under the stated curing and composition conditions. The problem emerges more generally when similar XRD or FTIR features are treated as if they alone prove a specific geopolymer backbone or a unique reaction pathway. Louati et al. [8] interpret XRD and IR observations in terms of phosphate phases and a proposed Si–O–Al–O–P network. Their Si/P–strength observations do not provide bond-specific validation; combining these measurements does not uniquely establish the network or reaction pathway. Dimas et al. [53] examined sodium-silicate-derived gels using XRD, FTIR, thermal analysis and microscopy; gel properties depended on silicate modulus and aluminium incorporation. This includes solid/gel characterization, not only solution-state observations. Its implications for identifying a unique hardened-binder network are assessed here as a methodological inference. In alkali-activated fly ash, Fernández-Jiménez and Palomo [54] related T–O band changes to gel amount, Si/Al and reaction time, with complementary XRD and chemical separation. The present review uses this combined evidence to motivate contextual interpretation of FTIR assignments. Phair and van Deventer [55] found that silicate activator pH altered silicate speciation and phase formation; calcium silicate hydrate present at pH 12 was largely unobserved at pH 14. Changes of this kind make it harder to infer phase–property relations from routine characterization alone. El Khomsi et al. [56] combined multinuclear MAS NMR, FTIR, and porosimetry to interpret network and pore-structure changes in metakaolin geopolymers with and without lithium tetraborate, illustrating the complementary roles of coordination-sensitive and microstructural evidence. McCaslin et al. [57] combined spatially resolved pair-distribution-function analysis with nano-XRF in carbonated alkali-activated slag, pairing atomic-structure information with elemental distributions as complementary evidence for phase interpretation. FTIR and XRD are not weak methods. Chemistry claims gain strength when their signals are integrated with other evidence and lose it when a single signal is treated as proof.

This limitation becomes more evident when analytical signals are asked to support chemistry-to-property explanations beyond their evidential reach. Studies using XRD, FTIR, and related methods frequently report broad associations between structural change and compressive strength, but the meaning of those associations remains system-dependent. Van Riessen et al. [58] reported separate XRD–strength correlations for fly-ash- and metakaolin-based binders and argued that the method was useful for both. These separate relationships do not demonstrate failed cross-system transfer; transferability beyond the tested formulations remains a question raised by the present review. Bernal et al. [59] used NMR to distinguish gel changes during accelerated carbonation in slag, fly-ash and blended binders. These are binder-dependent carbonation responses and do not directly compare contradictory chemistry–strength explanations. Walkley et al. [60] used multinuclear NMR to show that thermal treatment changed gel nanostructure across a series of slag and fly ash blends and related N-A-S-(H) formation to strength retention. Links of this kind rest on structure-sensitive evidence; routine bulk analysis alone would not have established them. García Lodeiro et al. [61] examined synthetic C–S–H gels with different additions of NaOH and reported changes in FTIR spectra together with enhanced carbonation at higher alkali contents. Their interpretation drew on FTIR, XRD, SEM/EDX, and selective dissolution, and considered both carbonation-related silicate polymerization and alkali-induced gel modification. Li et al. [62] examined strength and microstructure across activator formulations and fitted a GA-BP model. Those formulation-specific associations and model fits do not by themselves verify a gel mechanism. This review accordingly separates observed association and prediction from causal explanation.

Curing history and calcium-bearing additions affected performance differently across the systems examined. In Class F fly ash binders, Bakharev found that room-temperature precuring improved subsequent heat-cured strength development and that reaction-product composition depended on curing history [63]. In metakaolin geopolymers, Rovnaník reported that longer initial heat treatments increased early strength but reduced 28-day strength relative to ambient-cured specimens, alongside larger pores and increased cumulative pore volume [64]. Under room-temperature curing, Shamsah et al. reported increased reactivity within the first 24 h and higher compressive strength from three days onward with 5% slag substitution [65]. In a separate fly ash system, Temuujin et al. found that CaO or $Ca(OH)_2$ substitution increased seven-day compressive strength under ambient curing but decreased it under the tested 70 °C curing regime [66]. For the present diagnostic framework, these findings require strength trends to be interpreted within the tested formulation, curing history, and age; they do not establish a common reaction mechanism across systems. The same interpretive caution applies to calcium-bearing systems. In

metakaolin–slag blends, coexistence of geopolymeric gel and calcium silicate hydrate depends on alkalinity and reactive calcium availability [67]. Yip et al. explicitly presented the proposed gap-bridging contribution of coexisting geopolymeric and calcium silicate hydrate gels to strength as an interpretation for which they provided no experimental evidence [67]. In synthetic gel compatibility experiments, García-Lodeiro et al. examined precipitation/dissolution and direct mixing of pre-synthesised N-A-S-H with calcium silicate hydrate (C-S-H) or calcium aluminosilicate hydrate (C-A-S-H) gels. Gel stability and reaction depended on pH and calcium availability, and the authors predicted an equilibrium mixture of residual N-A-S-H and C-A-S-H [68]. These model-system findings do not directly demonstrate structurally mixed domains in actual blended binders. Across five low-calcium fly ashes, Gunasekara et al. related zeta potential to gel formation and compressive strength while also identifying workability and precursor physical characteristics as relevant factors [69]. Nath and Sarker found that increasing slag content in ambient-cured fly ash mixtures shortened setting time, reduced workability, and increased compressive strength [70]. A chemistry-to-property claim accordingly becomes stronger only when the study distinguishes among parallel contributors instead of assigning the observed performance trend to one simplified structural cause.

El Fadili et al. examined binary and ternary binders containing calcined clayey phosphate washing waste, fly ash and sugar lime at a fixed reported activator-to-solid mass ratio. Changing the solid blend from 60% phosphate waste/40% fly ash to 60% phosphate waste/32% fly ash/8% sugar lime increased the reported 28-day compressive strength from 45.73 to 47.50 MPa [14]. The replacement also changed the Ca, Si, Al and P inputs and precursor phase proportions. Their XRD, FTIR, SEM/EDS and thermal observations informed a mixed-gel interpretation, but did not separately quantify the proposed gel contributions or establish the proposed transition from diffusion-controlled to interfacial reaction-controlled hydration. The defensible result is therefore a formulation-specific performance change; its attribution to one calcium-mediated mechanism remains conditional [14].

In the fly ash–metakaolin series of Ben ali et al., increasing metakaolin replacement changed the reported ratios and the calcium, sulfate and other fly-ash-derived inputs together. The highest tested strength occurred at 30% replacement, whereas porosity continued to decrease at higher replacement levels [15]. XRD, FTIR and SEM/EDS observations supported discussion of reaction products and matrix development, but the proposed hybrid-gel assemblage and its specific contribution to strength were not separately resolved [15]. Neither this study nor the phosphate-waste study isolates a sulfate effect. Applying the framework therefore preserves the measured composition–property trends while bounding gel attribution, elemental causation and transferability to what the reported evidence discriminates.

Bakharev et al. [71] reported lower carbonation resistance for tested alkali-activated slag concrete than ordinary Portland cement (OPC) concrete under two exposure protocols. Efflorescence studies [72] address a separate durability outcome and cannot be combined with carbonation results to imply untested formulation–curing interactions. Neither outcome alone establishes a unique gel or mechanism.

Microstructural description can also be converted too quickly into mechanistic explanation. Fernández-Jiménez et al. [12] developed a descriptive model of microstructure evolution in thermally cured alkali-activated fly ash, showing partially reacted spheres, compacting gel, and later embedding of residual particles in a denser matrix. The authors proposed the descriptive sequence as a general model of fly-ash activation. In addition to microscopy and degree-of-reaction measurements, they cited earlier XRD, FTIR and NMR studies; those supporting data were not newly presented in the paper. However, SEM morphology does not by itself establish which dissolved species dominated, how condensation proceeded at the molecular scale, or whether a given gel assignment is unique. White et al. [40], in the introductory discussion of their multiscale simulation study, emphasize that dissolution and repolymerization occur concurrently, making experimental characterization of the individual processes challenging. The microstructure seen at one scale is the outcome of several overlapping processes, not a direct image of mechanism. When morphology is read as mechanism without intermediate evidential steps, the chemistry-to-structure logic becomes compressed beyond what the data can support.

Using ^{57}Fe Mössbauer spectroscopy, XRD, and SEM, Lemougna et al. [10] found largely unreacted ferroan forsterite but partial augite reaction in activated volcanic ash. New ferric environments suggested network participation or an amorphous gel-like phase; distorted tetrahedral and five-coordinated sites remained unresolved. Coordination-sensitive evidence thus distinguished phase-dependent reactivity while retaining uncertainty about the reaction product.

A structural indicator tracking a property does not establish singular causation. Koutník et al. [9] found that total water affected viscosity, mechanical performance, pore volume, and mean pore size, whereas alkali content did not significantly affect total pore volume and had a water-dependent viscosity effect. Those response-specific associations do not establish one molecular mechanism linking every property. Gel continuity, residual precursor, porosity, liquid chemistry, and impurities may contribute in parallel; alternative explanations require assessment.

These examples point to the need for an explicit evidence hierarchy in geopolymer chemistry, but not a single linear ladder. The strength of a chemistry claim depends on three partly independent questions. The first concerns observation strength: whether the signal is calibrated, reproducible and reported with its uncertainty and spatial or temporal resolution, and whether the measured specimen, sampled volume or observation interval adequately represents the material or process to which the claim refers. These five observation criteria are assessed jointly, without ordinal levels or an aggregate score. The second concerns causal inference, with correlation as the baseline and three complementary evidence pathways: controlled manipulation of the relevant variable, convergence of methods based on different physical principles, and explicit testing or exclusion of alternative explanations. These pathways are not successive ranks, and the presence of a pathway is not by itself proof of causation. Convergence provides independent support only to the extent that shared sensitivities, assumptions and sources of error have been assessed. Agreement between shifts of the FTIR Si–O–T band and the XRD amorphous hump should not be counted as independent confirmation merely because two instruments were used. Methods such as ^{27}Al NMR, selective dissolution and XRD likewise require an explicit account of their distinct evidential functions and shared limitations. Orthogonal convergence can strengthen structural inference but does not by itself establish causation. The third question concerns transferability: whether a finding is claimed only for the tested system, for a bounded system family, or across system families. Computational and modelling studies contribute evidence of a distinct type whose conditions must be stated, rather than occupying a fixed rank above experimental observation.

Because these dimensions are partly independent, a claim can be strong on one and weak on another. A well-resolved observation may support only a correlational property explanation, and a mechanism well established within one system may remain untested elsewhere. Gel identity, mechanism and property explanation should each be stated with the support reached on all three dimensions; the number of techniques reported does not settle the question.

The three dimensions provide the basis for Fig. 3. The hypothetical display records which observation criteria and causal evidence pathways are documented; it does not position a claim on ordinal observation or causality scales. Only transferability compares the supported and claimed scope, and broader scope is not equivalent to stronger evidence. No source is rated by the illustrative display.

Analytical results should be assessed against the claim they support. A chemistry–structure–property explanation requires calibrated observations, tests discriminating alternative explanations where causation is claimed, and a bounded transferability range. The presence of XRD,

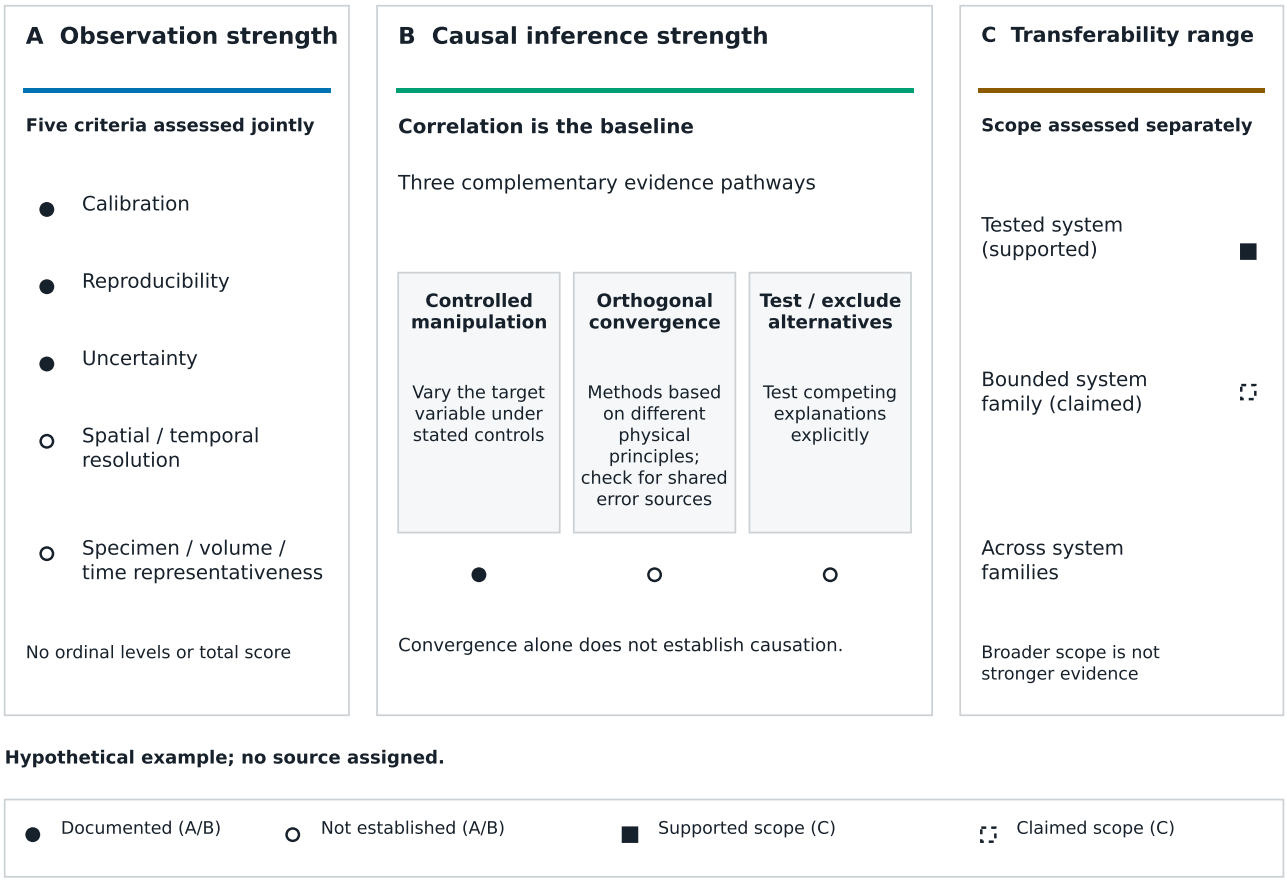


Fig. 3. Three partly independent dimensions of evidence for chemistry claims in geopolymer research. Observation strength is assessed through five jointly considered criteria: calibration, reproducibility, uncertainty, spatial or temporal resolution, and specimen, sampled-volume or observation-interval representativeness. Causal inference uses correlation as a baseline, with controlled manipulation, orthogonal convergence and testing or exclusion of alternatives as complementary evidence pathways, not successive ranks. Transferability distinguishes the tested system, a bounded system family and cross-family scope; broader scope does not imply stronger evidence. The hypothetical example assigns no rating to a source: filled and open circles in A and B indicate documented and not-established criteria or pathways, respectively; In C, a filled square indicates supported scope and a dashed-outline square indicates claimed scope. Designed for 190-mm double-column placement; do not reduce to single-column width. No total score is calculated, and orthogonal convergence alone does not establish causation.

FTIR, TG-DTA, or SEM alone does not meet those requirements.

Evidence-hierarchy discipline also benefits from advances in characterization resolution. Multi-nuclear solid-state nuclear magnetic resonance (^{27}Al, ^{29}Si, and where applicable ^{23}Na) provides direct information on aluminum coordination, silicon connectivity, and charge-balancing environments, as applied by Walkley et al. [60]. Routine XRD or FTIR alone does not resolve this information. Operando and synchrotron-based techniques, in turn, resolve time-dependent and spatially resolved phase evolution that single-timepoint laboratory measurements cannot isolate. Coordination-sensitive spectroscopies such as Mössbauer distinguish phase-dependent iron reactivity and changes in ferric environments that suggest possible network or amorphous gel participation [10]. Such evidence supports a more differentiated interpretation than a bulk-oxide balance alone, while leaving the precise structural location of the newly formed ferric environments unresolved. These methods do not automatically resolve interpretive overreach, but they give authors tools to escalate evidential support when conventional characterization leaves the chemistry underdetermined. Recent work on phosphate geopolymers illustrates this principle in practice: Keppert et al. [33] combined XRD, SAXS, FTIR, and solid-state NMR, and assigned each technique its own evidential function (crystalline phase identification, mesoscale structural organization, bond-environment fingerprinting and aluminum coordination, respectively) instead of treating the outputs as interchangeable confirmations of one claim. Such complementary multi-technique design, where each method addresses a distinct structural question, represents a positive model for evidence-proportional characterization practice.

9.1. *Evidence for specified chemical questions in representative studies*

The error domains discussed above can interact: weak system definition may accompany underdefined activator or precursor chemistry, while ratio ambiguity and limited analytical evidence can weaken mechanism or structure–property interpretation. The representative matrix examines selected claims and observations within individual studies so that each limitation can be located at the question it affects.

Table 1 illustrates bounded interpretations and claims requiring qualification, selected for system definition, reconstruction, and evidence discipline at the specified question.

A comparatively stronger reading position applies to a specified question when system identity, reconstruction of the relevant variables, and the evidence supporting the interpretation are sufficiently aligned. It does not imply that all conclusions in the study are equally supported or that its findings transfer to other binder chemistries.

A partially defensible reading position applies when useful observations support the specified question, but a remaining ambiguity in variable accounting or structural interpretation limits the conclusion. A narrow system scope, an application-oriented objective, or the use of several precursors is not sufficient on its own to justify this category.

A confounded reading position applies when unresolved ambiguities prevent the evidence from discriminating the specific chemical attribution being examined. The limitation must be identified at the level of

the claim, such as assigning particular gels from overlapping thermal responses without independent confirmation. Other observations in the same study may remain useful.

The positions apply to specified questions. Wang et al. [4] provides a comparatively stronger SAP/AAS synthesis, and Weng and Sagoe-Crentsil [5] a bounded low-Si/Al metakaolin interpretation. The confounded position for Burduhos Nergis et al. [11] concerns thermal gel assignments, not the number of precursors or all reported phase observations.

Cross-system comparison is assessed here for a specified claim, without placing materials in mutually exclusive classes. Fig. 4 asks the reader to document the activation route, calcium availability and other relevant constituents for each system before comparing ratio meanings, gel assignments or proposed mechanisms. Alkali, phosphate and mixed activation routes require explicit descriptions; a shared binder label does not establish chemical equivalence.

Calcium availability and the relevance of other constituents are described through reported amounts, calculation bases, phases and claim-relevant reactivity evidence. Bulk calcium content alone does not establish the calcium that participates in the reaction, and fly ash cannot be placed on a fixed side of a calcium division by its material name. Model precursors and industrial by-products may overlap in composition; a constituent is treated as relevant to an interpretation only when its identity and role in the specified question are stated. No universal calcium or impurity threshold is proposed.

The Fig. 4 decision tree checks whether the chemical meaning and calculation basis of the compared variables, relevant conditions and supporting evidence are sufficiently aligned for the specified claim. Missing information or an incompatible basis restricts the chemical attribution that can be transferred; descriptive contrasts may remain useful. Adequate specification permits a bounded comparison but does not by itself establish causality or cross-system transferability. This is an author-proposed interpretive aid, not a validated classification or quality score.

Table 1 presents twelve question-specific assessments using five fields: the claim reported by the source, the evidence used, the precise interpretive limitation, why a claim exceeds its evidence where applicable, and a defensible alternative. The source column identifies the publication, source passage and relevant section of this review. Bounded positive examples are retained as contrasts; an unsupported source-specific overextension is not invented to fill the diagnostic fields.

The reading positions are applied qualitatively using three criteria: whether the system family is stated; whether the variables relevant to the selected question can be reconstructed from the reported composition and procedure; and whether the analytical evidence supports the specificity of the chemical interpretation. Alignment across these criteria supports a comparatively stronger reading position. A bounded unresolved ambiguity supports a partially defensible position; ambiguities that prevent discrimination of the selected attribution support a confounded position. These positions remain interpretations of specified questions rather than overall study ratings. They are not additional table scores. The five fields separate the source's own claim from the present

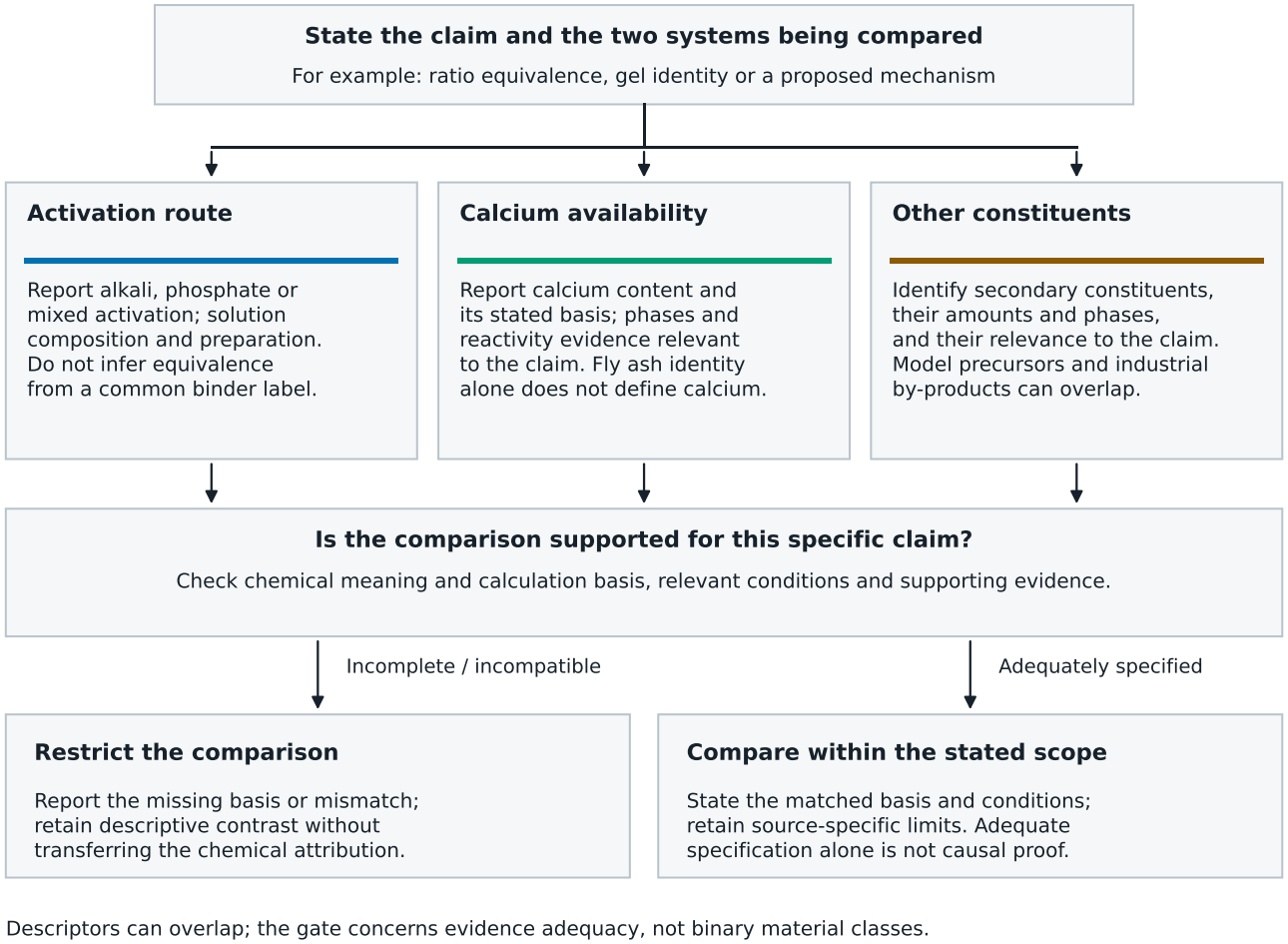


Fig. 4. Cross-system comparison for a specified claim. Author-proposed operational decision tree using three overlapping descriptors: activation route, calcium availability and other relevant constituents. These are not mutually exclusive material classes. The decision concerns the adequacy and compatibility of variable definitions, calculation bases, relevant conditions and supporting evidence. No universal calcium or impurity cutoff is imposed; material names do not assign fly ash or industrial by-products to fixed classes. Adequate specification alone does not establish causality or transferability. The tree has not been independently validated.

review's assessment of its support. System specificity and the use of multiple precursors are scope descriptors, not automatic reasons for an adverse classification. The assessment reflects the present author's reading and has not undergone independent inter-rater validation. It does not provide validated quality scores or prevalence estimates.

The distinction can be illustrated by three entries. Wang et al. [4] is comparatively stronger for distinguishing SAP and AAS system families; its role is comparative synthesis rather than experimental isolation of formulation variables. Louati et al. [8] reports a Si/P–strength relationship and XRD, IR, and SEM observations, but the proposed network structure and explanations involving dissolution and charge balance require qualification beyond those measured relationships. Its partially defensible position therefore concerns network and mechanism attribution, not the use of illito-kaolinitic clay or the absence of tests in other systems. Burduhos Nergis et al. [11] assigns N-A-S-H, C-S-H, and C-A-S-H contributions to thermal responses in the phosphate-activated formulation designated CAMK by the authors. Overlapping thermal ranges and the absence of independent confirmation of those gel identities limit that specific assignment. The confounded position concerns this inference; it does not invalidate the measured thermal features or the reported disappearance of the anhydrite diffraction signal in mine-tailings-containing samples.

10. A chemistry-focused corrective framework

The proposed corrective framework comprises six reporting and interpretation disciplines. It builds on definitional and mix-design concerns [2] and precursor, characterization, testing, and standardization issues [1], without prescribing a universal protocol. The six-part organization is developed in this review; those sources do not establish it independently.

The need for such discipline is not only conceptual but also operational. Shi et al. [73] discuss the need to examine activators, reaction mechanisms and product properties systematically. The present review extends that concern to explicit reporting for reconstruction and comparison. The foundational review by Duxson et al. emphasizes precursor characterization beyond bulk oxide composition and the importance of activator chemistry for dissolution and product development [74]. Broader methodological concerns regarding formulation and comparison are discussed in later review literature [75,76]. Interlaboratory studies support the same conclusion from a different direction. When mix design and mixing protocols were specified in advance, participating laboratories produced alkali-activated concretes with reproducible compressive strength rankings and deviations broadly comparable to those expected for Portland cement concrete [77,78]. In the companion durability programme, the influence of curing and test conditions was most evident in freeze–thaw results, whereas sulfate and alkali–silica reaction outcomes raised separate, method-specific questions [79]. These studies do not show that reproducibility fails without such specification; they show that it was attainable when the relevant formulation and processing variables were defined beforehand, which is the condition the present framework extends to chemistry reporting. Reproducibility and interpretation thus lead to the same requirement: chemistry has to be reported as information that can be reconstructed, not as a nominal description.

System identification discipline requires chemically explicit family boundaries before mechanism or variable interpretation. Low-calcium alkali-aluminosilicate, calcium-bearing, and phosphate-activated systems should remain distinguishable under any umbrella label. System identity bounds plausible pathways, structural expectations, and comparison sets.

Precursor definition discipline requires reactivity-relevant characterization beyond the material name: measurable silica and alumina availability, relevant amorphous or vitreous fraction, persistent crystalline phases, impurities, and activation history, including calcination temperature and duration where applicable. Detail should match the system and claim; naming does not replace characterization.

Activator definition discipline requires reconstructable solution chemistry. Alkali-system reporting includes hydroxide concentration, silicate modulus, cation, hydroxide/silicate amounts, water contribution, and soluble aluminum or other active species. Phosphate systems additionally require acid loading, dilution basis, relevant ratios, and phosphate- and aluminum-bearing species. Reagent names alone are insufficient.

Ratio definition discipline requires included components, calculation basis, total-batch or subset scope, and intended descriptive or interpretive use. Cross-study comparison requires justification of chemical comparability. Si/Al or Na/Al is interpretable only through the accounting that produced it.

Where conventional XRD, FTIR, SEM, and TG-DTA leave competing gel or pathway interpretations unresolved, coordination-sensitive spectroscopy, elemental mapping, advanced microscopy, or in-situ and synchrotron methods may supply relevant discrimination. The framework requires characterization matched to the claim, not routine use of every advanced technique.

Mechanism claim discipline distinguishes observed indicators, converging inferences, and hypotheses borrowed by analogy. Dissolution sequence, gel conversion, network reorganization, and bonding proposals should be bounded by system identity and evidence and labelled as demonstrated, supported, or interpretive.

The sixth element is evidence hierarchy discipline. Analytical tools should be evaluated by the specific observation or inference they support, rather than assigned a fixed rank. XRD, FTIR, SEM, TG-DTA, calorimetry or rheology can each provide valuable evidence, but none is self-validating proof of gel identity, bond formation or a full causal mechanism. Strong chemistry claims require calibrated, reproducible observations with claim-relevant uncertainty and resolution; causal claims additionally require controlled manipulation or independent, orthogonal convergence assessed against alternative explanations, rather than technique count alone. The claimed transferability range must be stated separately and bounded by the systems actually tested.

Table 4 consolidates the six disciplines into a reporting baseline. It identifies information needed before mechanism and property interpretation, with analytical requirements proportionate to the claim rather than a universal protocol.

The following application illustrates reconstruction of a nominal formulation; it does not validate the proposed framework.

Illustrative application. For a sodium silicate–sodium hydroxide activated metakaolin formulation, reconstruction requires the component amounts, commercial solution composition and its stated basis, density where volume is converted to mass, initial hydroxide concentration, molar silicate modulus, and water supplied by every liquid component. Total-batch, precursor-only and solution-only ratios should be distinguished explicitly. A reactive-basis ratio additionally requires element-specific evidence under stated activation conditions; an assumed value should be labelled as such. Quantitative XRD characterizes the phase inventory and amorphous fraction, while dissolution or reaction-extent measurements address different questions. These descriptors permit reconstruction of the nominal formulation and preparation history; they do not by themselves determine the evolving reactive environment, dissolved species or free hydroxide concentration. Supplementary Tables S1 and S2 compare the evidence roles and illustrate the accounting differences. Table 5 summarizes, for each diagnostic domain, the typical error, the recommended practice and where the corresponding reporting components and supporting material are detailed.

11. Limitations

Search record. The retained records do not contain a complete database-specific search and exclusion trail. The original searches cannot be independently reproduced, and the completeness of literature

Table 4
Claim-proportionate reporting baseline for chemistry-focused geopolymer studies.

Component	Minimum required information	Why it matters chemically	Common omission	Risk created by omission
System identity	Clear statement of system family: alkali-aluminosilicate, phosphate-activated, calcium-bearing alkali-activated, one-part, hybrid, etc.	Determines which mechanism, gel logic, and comparison set are chemically valid	Broad use of "geopolymer" alone	System misclassification
Precursor identity	Material name plus source and basic classification	Establishes material origin	Only generic label reported	Nominal similarity mistaken for chemical similarity
Precursor reactivity basis	Phase inventory and amorphous fraction; condition-specific dissolved Si/Al or reaction extent where measured, with method, conditions and uncertainty	Separates phase abundance from chemically available or reacted material	Amorphous content treated as reactive content	Unverified reactive input or conversion
Impurity chemistry	Major impurity-bearing phases or chemically relevant minor components	Impurities may alter gel chemistry, side reactions, or phase assemblage	Minor oxides ignored as inert	Mixed chemistry remains hidden
Thermal or pre-treatment history	Calcination temperature, duration, grinding, drying, or other activation history	Structural state of the precursor controls dissolution behavior	Material treated as unchanged by pre-treatment	Same precursor label hides different reactive states
Activator identity	Exact activator type(s), including hydroxide, silicate, acid, mixed route, dry activator, or added soluble species	The liquid phase defines the reactive environment	Only activator name reported	Activator cannot be chemically reconstructed
Activator composition	Component masses/volumes; supplier and lot; composition source and mass basis; SiO_2 and alkali-oxide contents; solids and water fractions; density at a stated temperature for volume-to-mass conversion; molar modulus; initial hydroxide concentration and basis	Allows nominal formulation and liquid contributions to be reconstructed	Only activator name or modulus	Different inventories appear equivalent
Activator preparation and storage	Addition sequence, mixing, preparation/use temperatures, time to use, storage and CO_2 exposure; observed precipitation or viscosity change; carbonate and method if measured, otherwise state not determined	Defines handling and distinguishes reported history from measured chemistry	Preparation history or measurement status omitted	Activator state cannot be assessed
Soluble reactive species in activator	Amounts and chemical forms of added Si, Al or phosphate species; accompanying ions, composition basis and total water contribution	Accounts for species supplied through the activating medium	Solution treated as chemically passive	Species inventory and stoichiometry misread
Ratio basis	Define atomic or oxide molar notation, included components and denominator; distinguish total, precursor-only, solution-only and condition-specific reactive bases; identify assumptions	Makes accounting and comparison explicit	Symbols without calculation basis	Non-equivalent ratios compared
Water accounting	Separate added water, water in each solution and total physical water; state water/precursor or water/dry-component denominator and any formal oxide-water convention	Distinguishes actual water inputs from bookkeeping conventions	One water value without source or denominator	Apparent agreement hides different water inputs
Curing context	Temperature, duration, sealed/open conditions, humidity, and sequence of curing stages	Chemistry and phase evolution are curing-sensitive	Only total curing time reported	Property differences are misattributed to composition alone
Analytical basis of chemistry claim	Link each method to a specific claim; report calibration, uncertainty, spatial/temporal resolution and specimen, volume or interval representativeness; assess shared errors, independent contribution and alternative explanations	Bounds observation and inference by their evidential support	Technique count alone	Apparent corroboration without independent support
Cross-study comparison basis	Explicit statement of why compared studies are chemically comparable	Fair comparison requires system alignment	Studies compared through broad labels only	False generalization across non-equivalent systems
Replication and variability	Separate independent batch/specimen and analytical replicate counts; state sampling unit and variability measure; justify replication relative to the claim	Assesses reliability of a chemical or property difference	Unspecified n or pseudo-replication	Uncertain reliability and exaggerated precision
Quantitative uncertainty	State uncertainty basis, fitting/deconvolution assumptions and relevant sensitivity checks; distinguish fit precision from sample and model uncertainty	Qualifies quantitative phase and coordination estimates	Fit quality treated as certainty	Unsupported quantitative specificity
Analytical quality control	Claim-relevant standards, calibration, blanks, recovery/reference checks, instrument stability and handling controls, as applicable	Checks measurement bias and artefacts	No measurement quality controls	Artefact misread as chemistry
Mass balance	For phase/reaction quantification, state material boundary and mass basis; account for residual precursor, amorphous, volatile/lost and unassigned fractions as relevant; report closure and uncertainty	Reconciles phase or reaction accounting	Unreported fractions or basis	Incompatible totals or conversion estimates
Charge balance	For structural formulae, state species, oxidation states, compensating ions and assumptions; identify unresolved imbalance	Checks internal consistency of the proposed structural model	No charge accounting	Unsupported structural formula

Entries are reporting recommendations applied in proportion to the intended claim. Preparation and composition examples are discussed in Section 6; unmeasured descriptors must not be presented as measured chemical states. Mass or charge closure and additional techniques do not alone demonstrate causation.

coverage cannot be established. Bibliographic identity and claim support were audited during revision, but a separate canonical-DOI search-coverage audit was not performed. The purposively selected studies support bounded diagnostic interpretation, not field-wide frequency estimates.

Domain derivation. The available records do not establish the original derivation process for the six domains, including a mixed deductive–inductive procedure or a reduction from eight candidate domains to six. The claim-specific coding unit and overlap rule were applied during revision to the twelve Table 1 entries; their use throughout the original search or literature set is not established. The domains form a conceptual diagnostic organization rather than an empirically validated literature classification.

Independent assessment. The claim-specific assessments in Table 1 were made by the present author and have not undergone independent second assessment or inter-rater validation. They are diagnostic

Table 5
Correspondence between diagnostic domains, typical errors and recommended practice. The table summarizes recommendations developed in Sections 4–10; it is not a universal protocol or a validated quality score.

Diagnostic domain	Typical error	Recommended practice	Table 4 components	Where detailed
Terminology and system definition (L1)	Umbrella term ("geopolymer", "alkali-activated material") used without a stated system boundary	State the system family (low-calcium alkali-aluminosilicate, calcium-bearing, phosphate-activated, one-part or hybrid); qualify "geopolymer" when the formulation departs from the lower-impurity alkali-aluminosilicate model	System identity; Cross-study comparison basis	Section 4; Table 3; Fig. 4
Precursor chemistry and reactivity (L2a)	Material name or bulk oxide list treated as reactive input	Report the phase inventory and amorphous fraction separately from measured dissolution or reaction extent, with the conditions of each measurement; report impurities and thermal or other pre-treatment history	Precursor identity; Precursor reactivity basis; Impurity chemistry; Thermal or pre-treatment history	Section 5; Supplementary Table S1
Activator chemistry (L2b)	Activator reported by name or modulus only	Report component amounts and composition basis, cation, hydroxide and silicate contributions, water supplied by each liquid, added soluble species, and preparation and storage history	Activator identity; Activator composition; Activator preparation and storage; Soluble reactive species in activator	Section 6; Section 10 (illustrative application)
Composition ratios and chemical variables (L3)	Ratio symbols compared without notation or calculation basis	State atomic or oxide notation, included components, denominator and basis (total batch, precursor-only, solution-only or reactive); label assumed reactive fractions; compare only chemically comparable systems	Ratio basis; Water accounting; Cross-study comparison basis	Section 7; Supplementary Table S2
Mechanism interpretation (L4)	Mechanism inferred from rate data or structural resemblance, or transferred from another system	Distinguish observed indicators, converging inferences and analogy; label claims as demonstrated, supported or interpretive; bound them by system identity and tested conditions	Analytical basis of chemistry claim; Curing context	Section 8; Fig. 3
Chemistry–structure–property interpretation (L5)	A single analytical signal treated as proof of gel identity or of a causal mechanism	Link each method to a specific claim; report calibration, uncertainty, resolution and representativeness; assess alternative explanations; do not count related signals as independent confirmation; state the transferability range	Analytical basis of chemistry claim; Replication and variability; Quantitative uncertainty; Analytical quality control; Mass balance; Charge balance	Section 9; Table 1; Fig. 3

interpretations of the specified passages and evidence, not validated scores of overall publication quality. Agreement between assessors and classification reliability have not been established.

Comparator selection. Table 2 compares six selected reviews. A documented comparator-selection trail and an exhaustive search for related reporting frameworks are not available. The comparison supports a diagnostic reorganization relative to these six reviews, not a field-wide claim of unique novelty or absence of related frameworks.

Chemical scope. System families differ in precursor composition, activation route, product assemblage and analytical visibility [4]. A claim may involve more than one diagnostic domain. The categories are tools for specifying the question and its dependencies, rather than mutually exclusive classes of publications.

Experimental-design scope. Chemistry-specific interpretation includes claim-proportionate reporting of replication, variability, quantitative uncertainty and analytical quality control when chemistry-to-property causation is proposed. The review does not prescribe a universal replicate count, conduct retrospective statistical reanalysis, or provide a complete general experimental-design standard. Broader statistical design and long-term durability questions are addressed only where they intersect with chemical interpretation.

Verification scope. Bibliographic identity and passage-level support checks do not establish complete correction or retraction screening. For eleven DOI records, Crossref metadata and the Retraction Watch database were screened and no linked notice was found. Crossmark data could not be obtained in this check, and publisher update histories could not be inspected; publisher-level verification was therefore not achieved. Missing source-discovery information is not reconstructed from verification provenance. The completeness of original discovery provenance and of publisher-level publication-status screening has not been established. Of the 79 retained publications, 67 checks are recorded as complete by their stated methods (62 Crossmark and five publisher-page/Crossref checks), eleven records have only record-level screening, and Reference [3] remains separately qualified. The two added experimental sources were checked through their Crossmark records and Crossref metadata; their publisher-page update histories could not be inspected.

These limitations define the scope of the review. Its contribution is a conceptual diagnostic clarification of selected chemistry-related interpretive and reporting risks.

12. Conclusions

The selected evidence illustrates interpretive risks arising from unstable terminology, insufficient precursor specification, underdefined activators, context-free ratios, simplified mechanism narratives, and overextended chemistry–structure–property claims. The proposed diagnostic framework locates those risks at specified questions and distinguishes observation strength, causal inference, and transferability. Its use does not establish field-wide error frequencies, validated quality scores, or independent framework validation.

Four practical consequences follow. A shared precursor class does not guarantee the same reactive input, and a shared activator label does not guarantee the same solution chemistry. Equal Si/Al values can describe different reaction environments, and the same nominal chemistry can be reached through different causal mechanisms.

Explicit system definitions, reconstructable formulations, and evidence-proportional interpretation are proposed to support reproducibility and defensible comparison. The reporting baseline and worked accounting example provide practical ways to state what can be reconstructed and what remains unresolved. Data-driven terminology screening, reporting checks, and evidence integration may assist that work, provided they retain chemical judgment and explicit verification limits.

CRediT authorship contribution statement

Kemal Şahbudak: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization, Data curation, Formal analysis.

Ethics statement

This review article does not contain any studies with human participants or animals performed by the author.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used generative AI and AI-assisted tools to support literature searching, language editing, consistency checking and the comparison of cited statements with source documents. All outputs of these tools were treated as proposals. At the decision stage, the author read the relevant source documents and made all decisions on content, citations and interpretation. The author reviewed and edited the manuscript as needed and takes full responsibility for the content of the publication.

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The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.mtcomm.2026.116442.

Data availability

All information supporting this review is contained within the article and its supplementary materials. No separate research dataset was generated.

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